Exercises in Marine Ecological Genetics

07. Genetic clustering

- Principal component analysis (PCA)
- Admixture analysis
- Complex plots with ggplot

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https://github.com/mhelmkampf/meg26

Today's objectives

By the end of this session, you will be able to:

- Understand the principles of genetic clustering and its role in **exploring relatedness and population structure**
- Perform **principal component and admixture analyses** on SNP data, including data preparation
- **Interpret the corresponding results** and recognize potential sources of bias
- Create **publication-quality plots** of PCA and admixture results in R

Principal component analysis

- **Reduces thousands of SNPs** into a small number of principal components that capture the most genetic variation among individuals
- Treats **each SNP as a variable** and encodes genotypes numerically (e.g. individuals × SNPs; 0 = AA, 1 = AB, 2 = BB)
- Calculates **covariance matrix** (individuals × individuals) and extracts PCs via eigenvalue decomposition
- PC1 explains the most variance, PC2 the next most, and so on
- Plots individuals (= dots) based on PC scores in **2D space**

▸ Reveals relationship between individuals (relatedness), population structure, and outliers, but does not assign individuals to groups

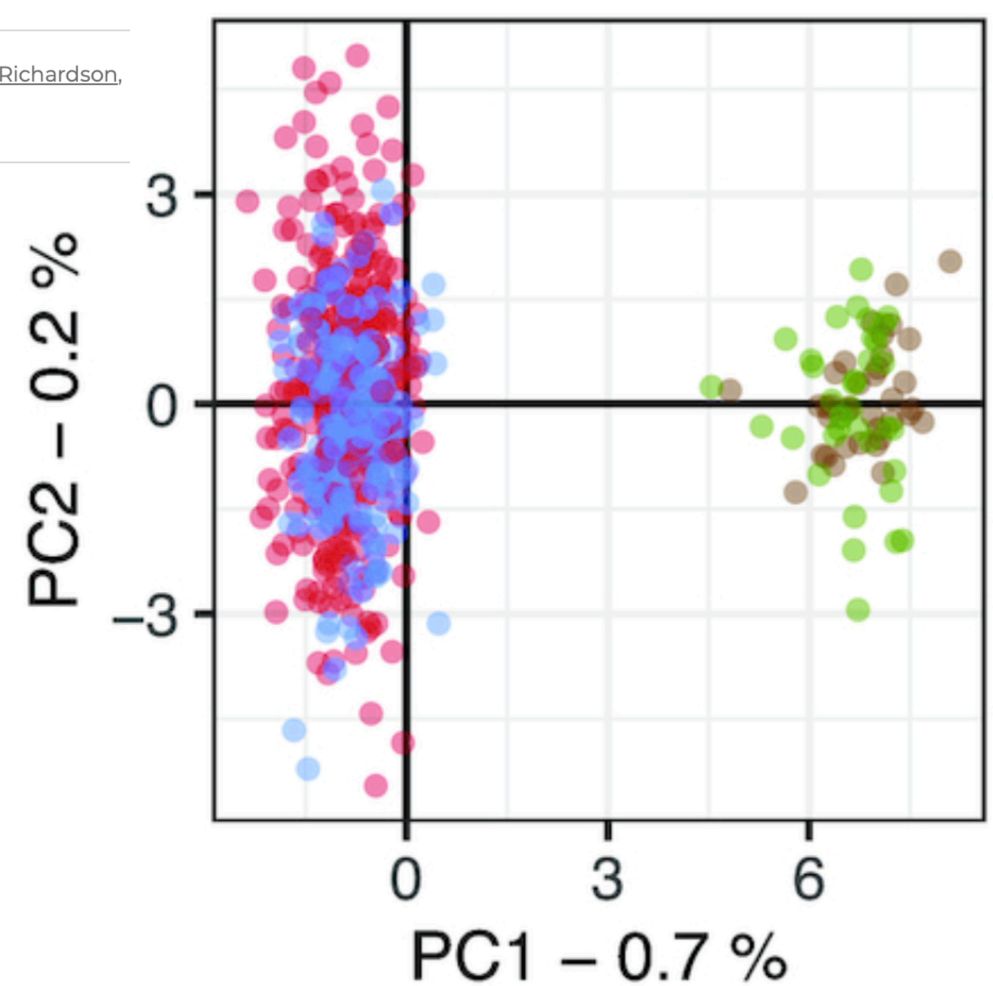
Principal component analysis

Example

Quantifying genetic differentiation and population assignment between two contingents of Atlantic mackerel (*Scomber scombrus*) in the Northwest Atlantic

Authors: Audrey Bourret, Andrew Smith, Elisabeth Van Beveren, Stéphane Plourde, Kiersten L. Curti, Teunis Jansen, David E. Richardson, Martin Castonguay, Naiara Rodriguez-Ezpeleta, and Geneviève J. Parent | AUTHORS INFO & AFFILIATIONS

Publication: Canadian Journal of Fisheries and Aquatic Sciences • 23 March 2023 • https://doi.org/10.1139/cjfas-2022-0232



NEA – Bay of Biscay NEA – Greenland

NWA – Canada NWA – US

> 10,000 SNPs

Admixture analysis

- **Estimates ancestry proportions** for each individual by assuming *k* ancestral populations (typically across a range, e.g. k = 2–8), based on genome-wide SNPs
- Uses **model-based clustering** to estimate ancestral allele frequencies (P-values), and which SNPs likely came from which ancestral population
- Calculates **ancestry proportions** (Q-values) for each individual and iteratively refines them, together with P-values, to best explain the observed SNP genotypes (e.g., 60% ancestry from population 1, 40% from population 2)
- Visualizes these ancestry proportions as **stacked bar plots**, with one bar per individual

▸ Reveals admixture (mixed ancestry) and population structure; can assign individuals to populations when *k* is well-supported

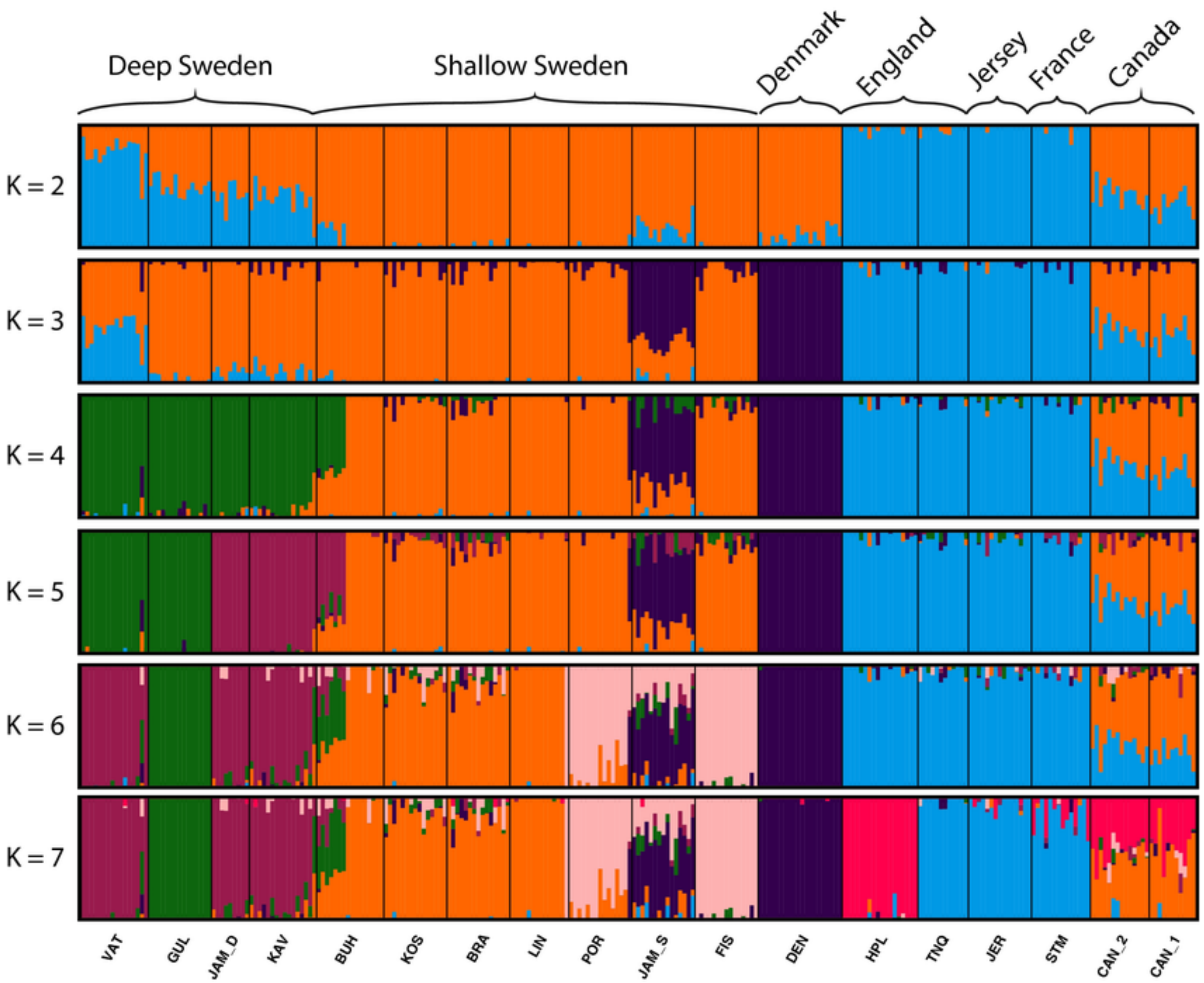
Admixture analysis

› Evol Appl. 2019 Dec 3;13(3):600-612. doi: 10.1111/eva.12893. eCollection 2020 Mar.

Secondary contacts and genetic admixture shape colonization by an amphiatlantic epibenthic invertebrate

Jamie Hudson [1], Kerstin Johannesson [2], Christopher D McQuaid [3], Marc Rius [1 4]

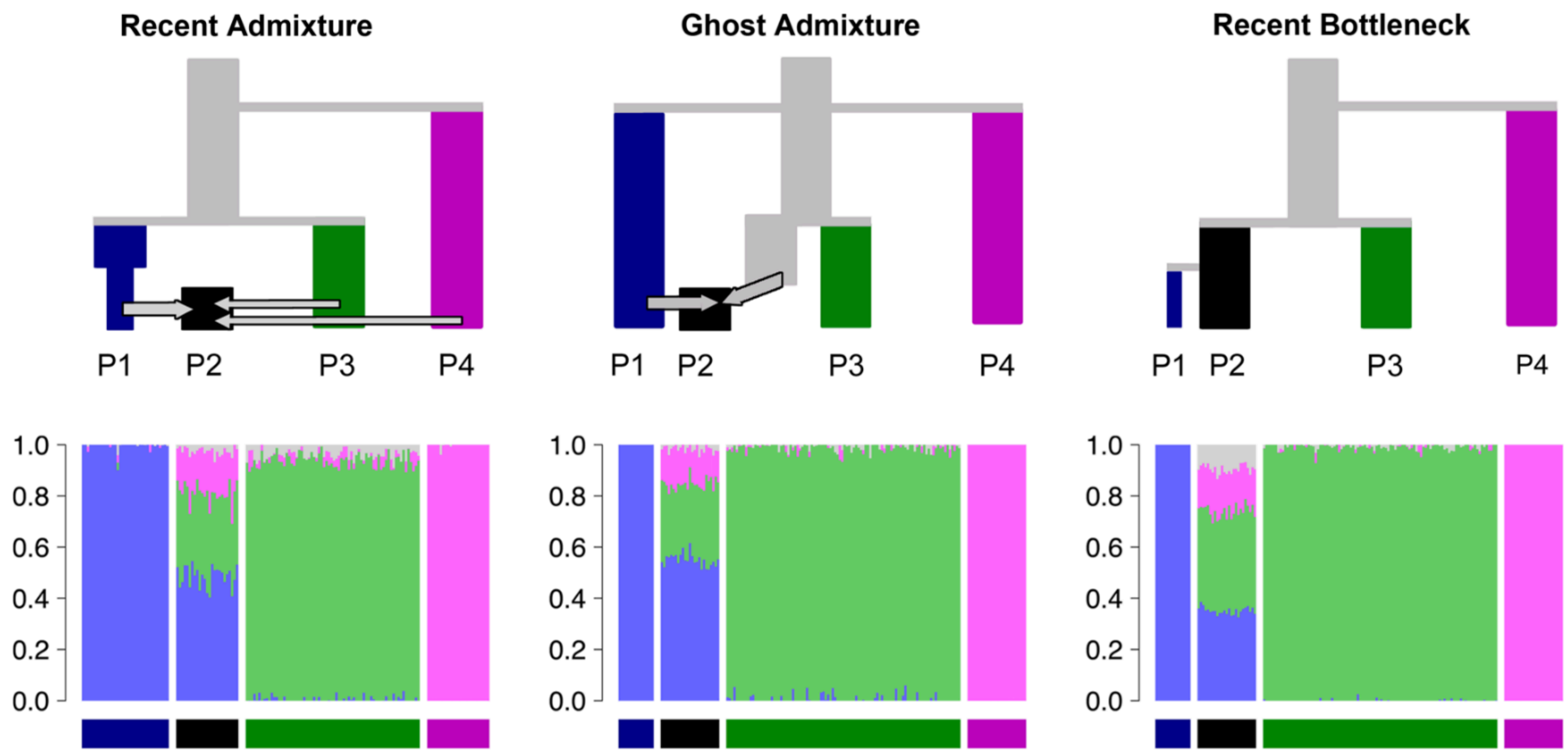
Example



Ciona intestinalis

~ 1600 SNPs

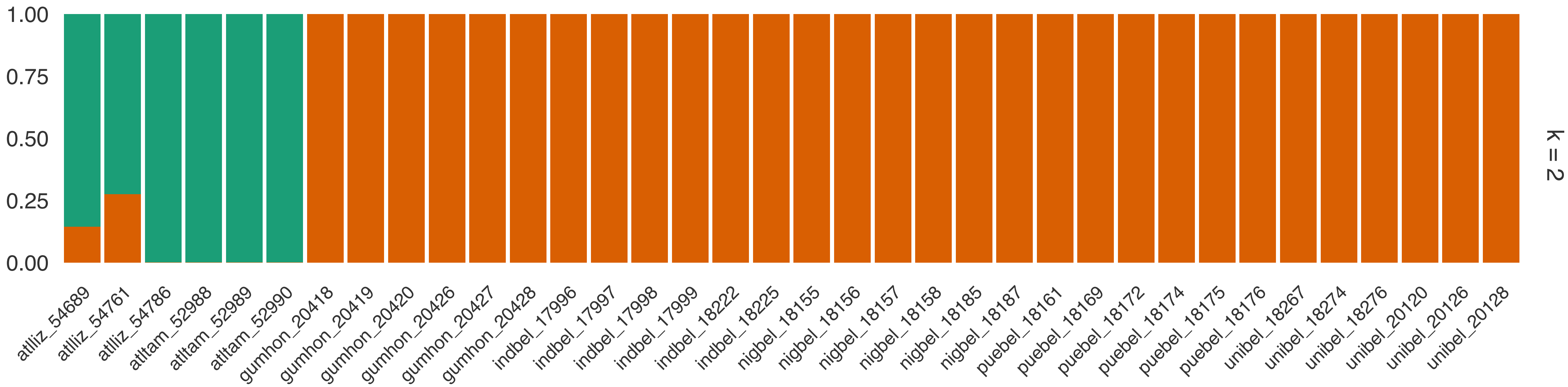
Limitations

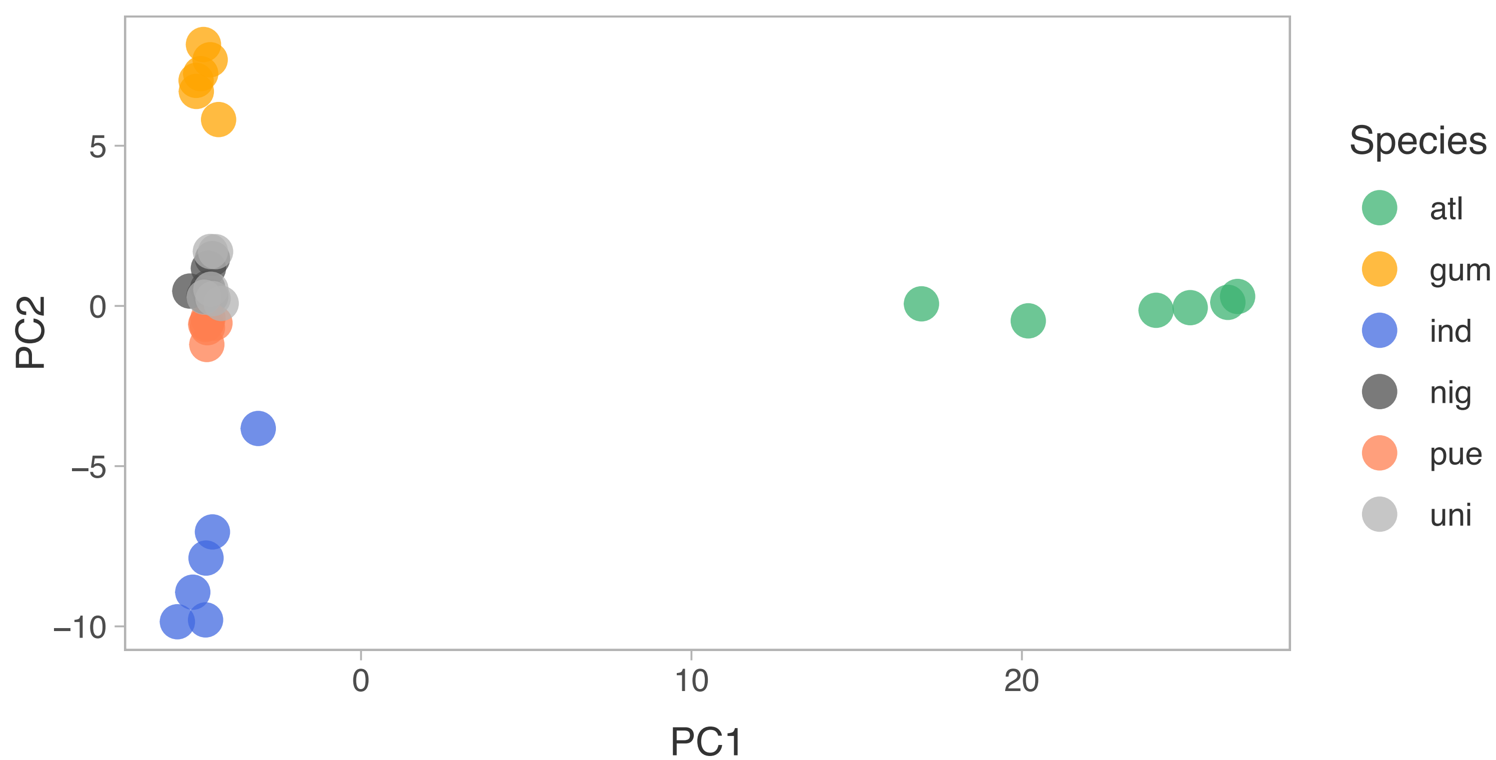


- Demographic histories other than recent admixture can lead to the same results
- Populations are assumed to be in HWE

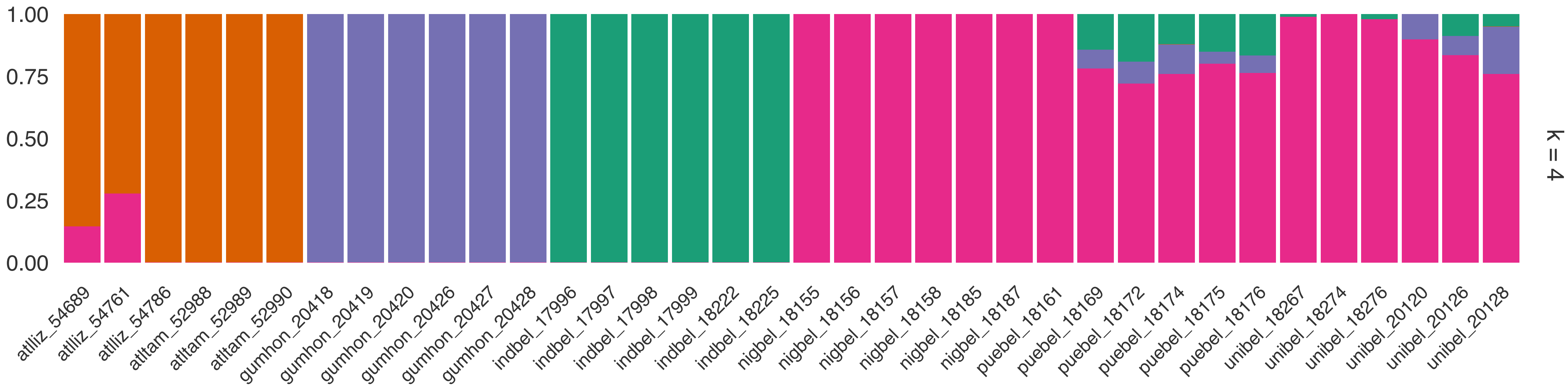
Lawson et al. 2018, *Nature Communications*

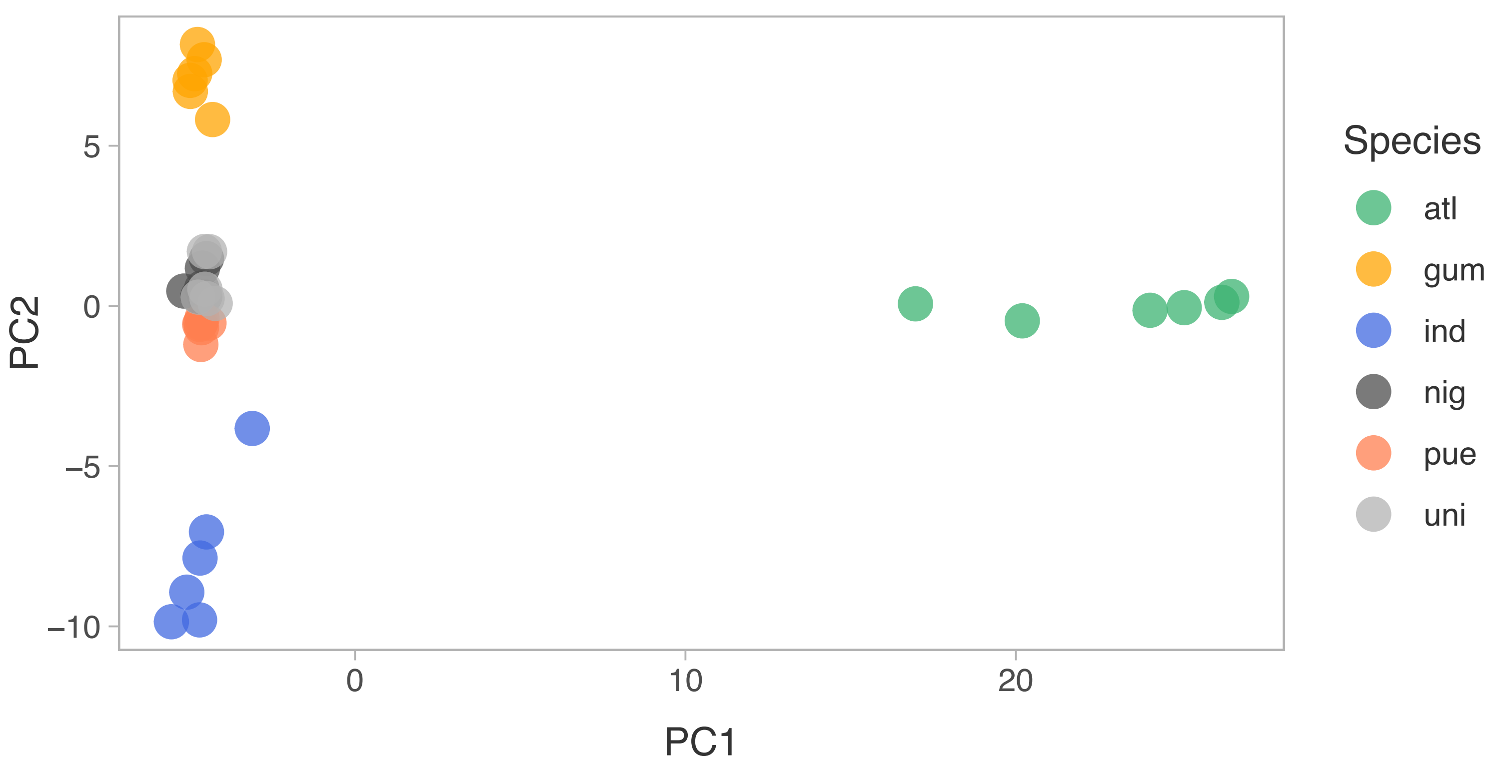
Admixture and PCA compared





Admixture and PCA compared

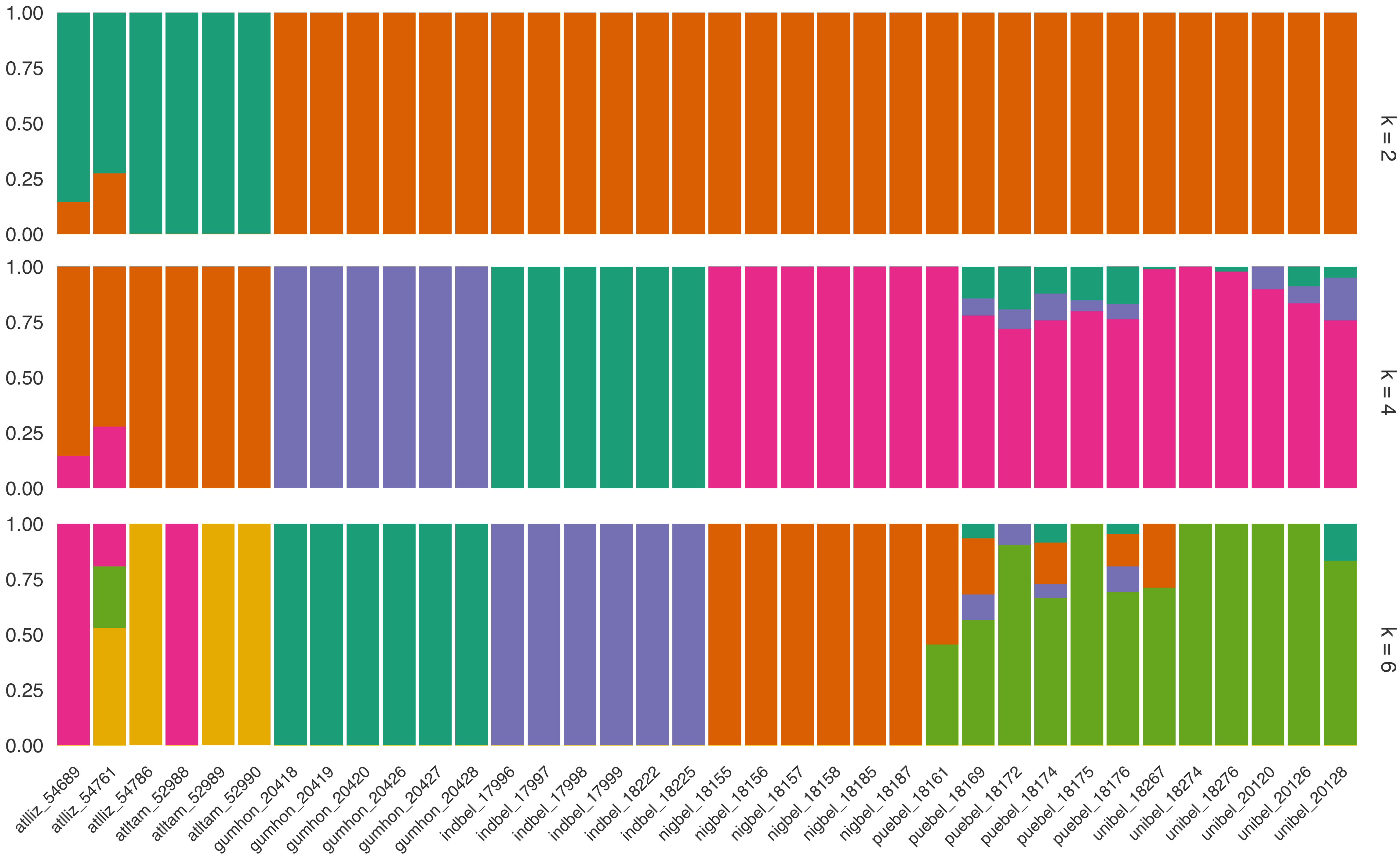




Summer 2026

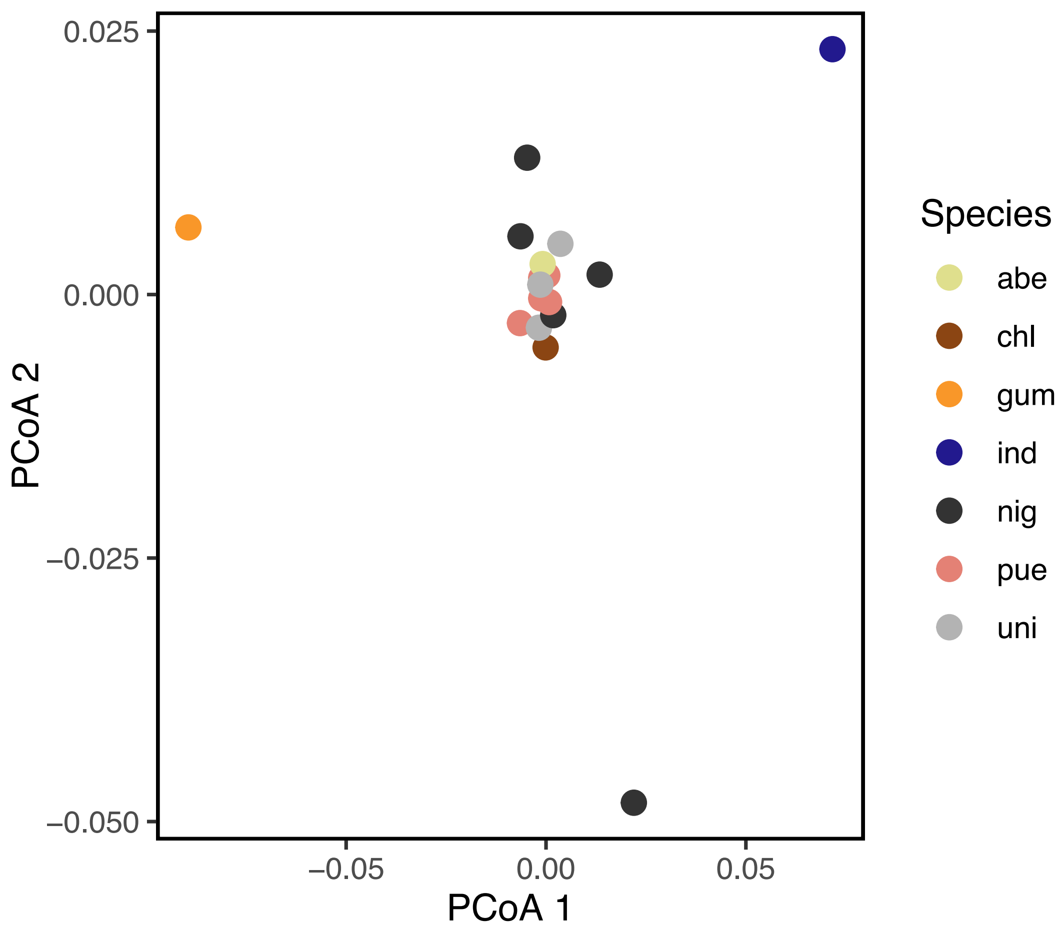
Carl von Ossietzky Universität Oldenburg

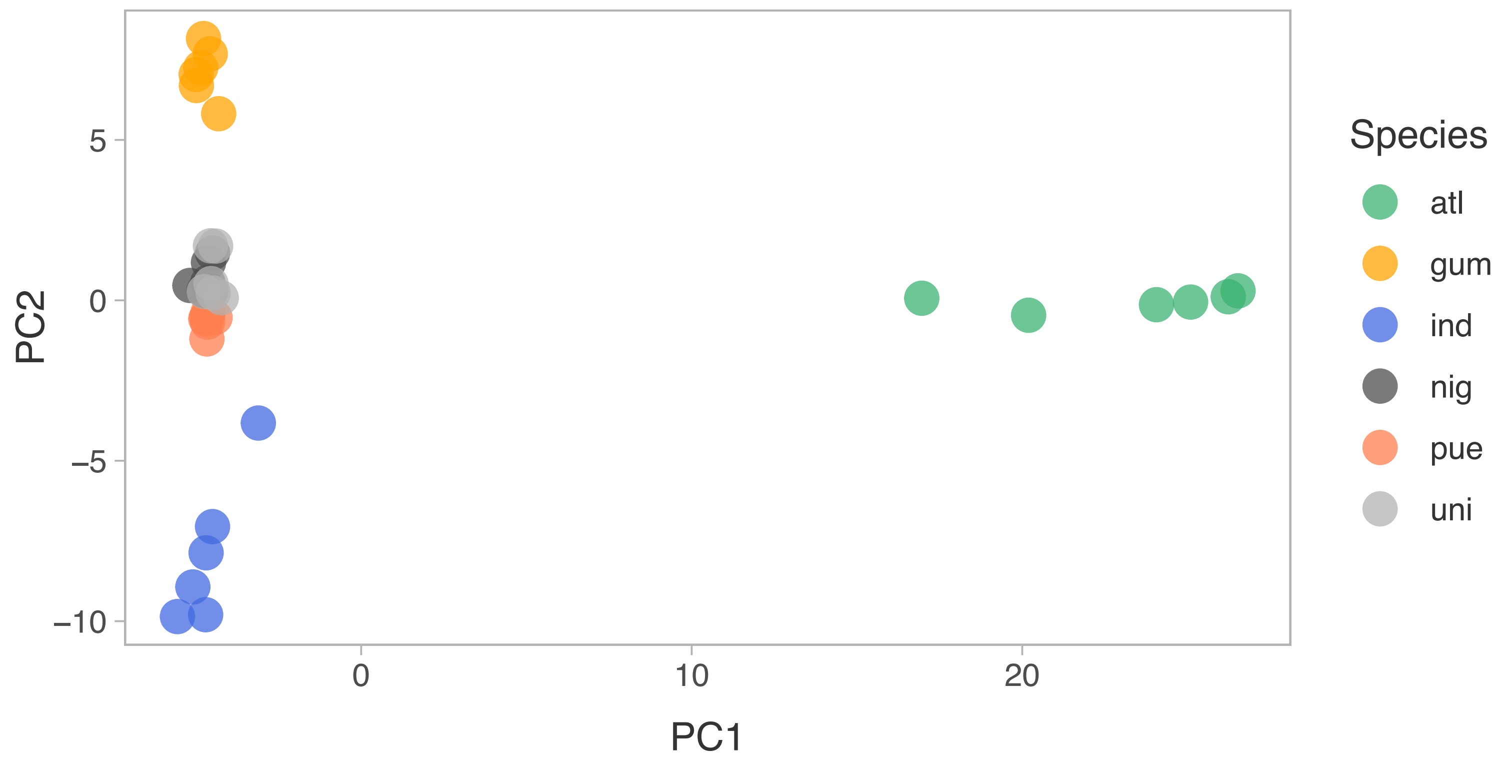
Comparing different *k*



Pairwise F_{ST} PCoA compared to PCA

- F_{ST} PCoA was based on few microsatellites, not genome-wide SNPs
- Shows per-subpopulation, as opposed to per-individual, patterns





Genetic clustering

Summary

```
gl <- vcfR2genlight(read.vcfR(“vcf_file”))                # Read VCF file into R
                                                 # and convert to genlight object

glPca(gl, nf = 2)                              # Calculate first 2 principal components

ggplot(tbl, aes(x = var1, y = var2)) +  # Plot with ggplot (tbl = tibble object)
  geom_bar() +
  labs() +
  guides() +
  theme()

### <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< bash >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>

for k in {2..8}                                    # Run ADMIXTURE analysis for k = 2–8
do
    admixture \
    --cv \
    file.bed $k > k${k}.out
done
```

Key takeaways

Summary

- Genetic clustering of SNP data reveals relatedness, population structure, and potential outliers **without requiring prior population labels**
- PCA reduces high-dimensional SNP variation to a few principal axes, enabling r**apid visualization of genetic clusters**
- Admixture analysis estimates the proportion of each individual's genome derived from hypothetical ancestral populations, revealing **patterns of mixed ancestry**
- Both methods are complementary and require **careful data preparation and interpretation**, with results influenced by factors such as linkage disequilibrium and model parameters (e.g., *k*)